Report

Of

Data Analytics with Python & Power BI

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 2410030691

Name of Student: KUNAL AGRATAM

Batch: 2024-28

CANDIDATE'S DECLARATION

I, Kunal Agratam, do hereby declare that the internship report titled "**Data Analytics with Python & Power BI**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: KUNAL AGRATAM
Roll No: 2410030691

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to sincerely thank EduSkills Academy for providing me the opportunity to undertake the 8-week internship in Data Analytics with Python & Power BI, and for the structured curriculum, mentorship, and real-world projects that helped me build practical skills in this domain. I also thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout this internship.

Student Name: KUNAL AGRATAM
Roll No: 2410030691

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INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE OF INTERNSHIP

EduSkills ACADEMY

PRESENTED TO

Kunal Agratam

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Data Analytics with Python & Power BI

During the internship, the learner demonstrated proficiency in:

- Welcome to Data Analytics & Environment Setup & Core Python for Data Analysis
- Data Cleaning & Preprocessing with Pandas & SQL Fundamentals for Data Analysis
- Advanced SQL Techniques & Data Extraction & Data Visualization with Power BI - Basics
- Python & SQL Integration for Data Extraction & Power BI Data Modeling & DAX Basics
- Advanced Power BI Visualizations & Interactivity & Python for Data Visualization
- Python & Power BI Integration & Power BI Service & Collaboration
- Real-World Project: Sales Performance Analysis & Real-World Project: Customer Behavior Analysis & Best Practices & Project Workflow
- Capstone Project: E-commerce Sales & Customer Insights Dashboard

This certification recognizes the successful completion of the internship requirements.

Issued on: Jul 28, 2026
Certificate ID: 2026-9ACA033384

Mitu Swain
General Manager
(Learning & Development)
EduSkills

CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents an account of the 8-week internship undertaken in the domain of Data Analytics with Python & Power BI, conducted by EduSkills Academy. Data analytics has become a critical function for organizations that wish to convert raw, unstructured data into actionable business insight. The internship was designed to provide a hands-on, project-based understanding of the complete data analytics workflow — from data collection and cleaning, through exploratory analysis in Python, to interactive dashboarding and reporting in Microsoft Power BI. The certificate of completion for this internship (Certificate ID: 2026-9ACA033384, issued on 28 July 2026) is attached in Chapter 3 of this report.

4.2 Organization Profile

EduSkills Academy is a skilling and industry-academia collaboration initiative that works to prepare a digitally skilled, Industry 4.0-ready workforce among students in India. It partners with academic institutions to deliver structured, certification-based internship programs across emerging technology domains such as data analytics, cloud computing, artificial intelligence, and cybersecurity, with the objective of bridging the gap between classroom learning and industry requirements.

4.3 Problem Statement

Organizations today generate large volumes of transactional and behavioural data but often lack the in-house capability to clean, analyze, and visualize this data in a way that supports timely decision-making. Raw data stored across spreadsheets, databases, and business applications remains underutilized unless it is systematically processed and presented through meaningful visual reports. The internship addressed this problem by training interns to build an end-to-end analytics pipeline — using Python for data preparation and SQL for structured data extraction — and to present the resulting insights through interactive Power BI dashboards that support business decision-making.

4.4 Project Objectives

- To set up and use a Python-based environment for data analysis, including core Python programming concepts.
- To clean, preprocess, and structure raw datasets using the Pandas library.

- To learn and apply SQL for data extraction, querying, and integration with Python.
- To design interactive dashboards and reports using Power BI, including data modeling and DAX.
- To integrate Python and Power BI for advanced data visualization and automated reporting.
- To apply the above skills on real-world style projects, culminating in a capstone dashboard.

4.5 Scope of the Project

The scope of the internship covered the complete lifecycle of a data analytics project: environment setup, data cleaning and preprocessing, database querying using SQL, exploratory and visual data analysis using Python, and business intelligence dashboarding using Power BI. The internship concluded with two real-world style projects — a Sales Performance Analysis and a Customer Behaviour Analysis — followed by a capstone project on an E-commerce Sales & Customer Insights Dashboard, which consolidated all the skills learned over the eight weeks.

4.6 Technologies and Tools Used

- Programming Language: Python (including the Pandas library for data manipulation)
- Database Query Language: SQL (fundamentals to advanced techniques and data extraction)
- Business Intelligence Tool: Microsoft Power BI (data modeling, DAX, visualizations, Power BI Service)
- Integration: Python–SQL integration and Python–Power BI integration for automated workflows

4.7 System Architecture

The analytics workflow followed a layered architecture. At the data layer, raw data was sourced and queried using SQL. At the processing layer, Python and the Pandas library were used to clean, transform, and structure the data. At the modeling layer, the processed data was imported into Power BI, where relationships were established and DAX measures were created. At the presentation layer, interactive visualizations, dashboards, and reports were built and published through the Power BI Service for collaboration and sharing.

4.8 Methodology

The internship followed a structured, week-wise curriculum in which theoretical concepts were reinforced through hands-on exercises and progressively complex projects. The week-wise breakup of topics covered, as reflected in the internship completion certificate, is summarized below:

Table 4.1: Week-wise Breakup of Topics Covered

Week	Topics Covered
Week 1	Welcome to Data Analytics & Environment Setup; Core Python for Data Analysis
Week 2	Data Cleaning & Preprocessing with Pandas; SQL Fundamentals for Data Analysis
Week 3	Advanced SQL Techniques & Data Extraction; Data Visualization with Power BI - Basics
Week 4	Python & SQL Integration for Data Extraction; Power BI Data Modeling & DAX Basics
Week 5	Advanced Power BI Visualizations & Interactivity; Python for Data Visualization
Week 6	Python & Power BI Integration; Power BI Service & Collaboration
Week 7	Real-World Project: Sales Performance Analysis; Real-World Project: Customer Behaviour Analysis; Best Practices & Project Workflow
Week 8	Capstone Project: E-commerce Sales & Customer Insights Dashboard

This methodology ensured that foundational concepts (Python, Pandas, SQL) were established first, followed by visualization and modeling concepts in Power BI, and finally applied through real-world and capstone projects that simulated actual business analytics scenarios.

4.9 Expected Outcomes

- Ability to independently clean, preprocess, and analyze raw datasets using Python and Pandas.
- Proficiency in writing SQL queries for data extraction and integration with analytical tools.
- Capability to build data models and interactive dashboards using Power BI and DAX.
- Practical experience in analyzing sales performance and customer behaviour data.
- A completed capstone project — an E-commerce Sales & Customer Insights Dashboard — demonstrating end-to-end analytics competency.

4.10 Certificates of Completion and Other Communication Proof

The Internship Completion Certificate issued by EduSkills Academy (Certificate ID: 2026-9ACA033384, issued on 28 July 2026, signed by Mitu Swain, General Manager

– Learning & Development, EduSkills) has been attached in Chapter 3 of this report as proof of successful completion of the internship requirements.

CHAPTER 5: BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, "Data Analytics with Python & Power BI" Internship Certification Program, 2026.
2. Microsoft, Power BI Documentation, https://learn.microsoft.com/power-bi/
3. Python Software Foundation, Python 3 Documentation, https://docs.python.org/3/
4. Pandas Development Team, Pandas Documentation, https://pandas.pydata.org/docs/
5. Internship Completion Certificate, Certificate ID: 2026-9ACA033384, EduSkills Academy, issued 28 July 2026.